VLAN

A **Virtual Local Area Network (VLAN)** is a logical network configuration that divides a physical network into multiple separate networks or broadcast domains. Devices connected to the same physical switch can be grouped into different VLANs based on their function, department, or network requirements, regardless of their physical location. Each VLAN is identified using a unique VLAN number and can be assigned a specific name for easier management.

Devices within the same VLAN can communicate with each other as part of the same logical network, while communication between different VLANs requires routing through a router or Layer 3 switch. VLANs help reduce unnecessary broadcast traffic and provide better network organization, security, and performance by separating devices into logical groups.

VLANs are widely used in modern Ethernet networks because they provide flexibility and make network management easier. They are commonly used in **schools, offices, universities, computer laboratories, and other Local Area Networks (LANs)** to separate users, departments, or services while using the same physical network infrastructure.

Advantages of VLAN

- Improves network security by logically separating devices into different VLANs and limiting unnecessary communication between them.
- Reduces broadcast traffic because broadcasts are limited to devices within the same VLAN.
- Provides better network management by allowing devices to be grouped logically regardless of their physical location.
- Improves network performance by reducing the size of broadcast domains and unnecessary network traffic.
- Easy to manage and reorganize because devices can be moved between VLANs through switch configuration without changing the physical network connections.
- Provides better flexibility by allowing different departments, groups, or types of users to be placed in separate logical networks.

Disadvantages of VLAN

- More complex to configure because VLANs require proper configuration of switch ports, VLAN IDs, and related network settings.
- Requires additional networking knowledge to configure VLANs, trunk ports, and inter-VLAN routing correctly.
- Communication between different VLANs requires routing, usually through a router or Layer 3 switch.
- Configuration errors can affect network connectivity and may prevent devices from communicating properly.
- A failure of the router or Layer 3 device can affect communication between different VLANs.
- Managing multiple VLANs can become complicated as the size of the network increases.
- Trunking and inter-VLAN routing may require additional configuration, increasing the overall complexity of the network.

Simulation / Design in Cisco Packet Tracer

This section presents the **VLAN network designed and configured in Cisco Packet Tracer**. The network consists of multiple end devices connected to a switch, with selected devices grouped into **VLAN 20 (STUD)** to provide logical network segmentation. The switch is connected to a router through a trunk link, allowing VLAN traffic to be managed and routed between the connected networks. This arrangement represents the structure and working of a **Virtual Local Area Network (VLAN)**.

Network Diagram

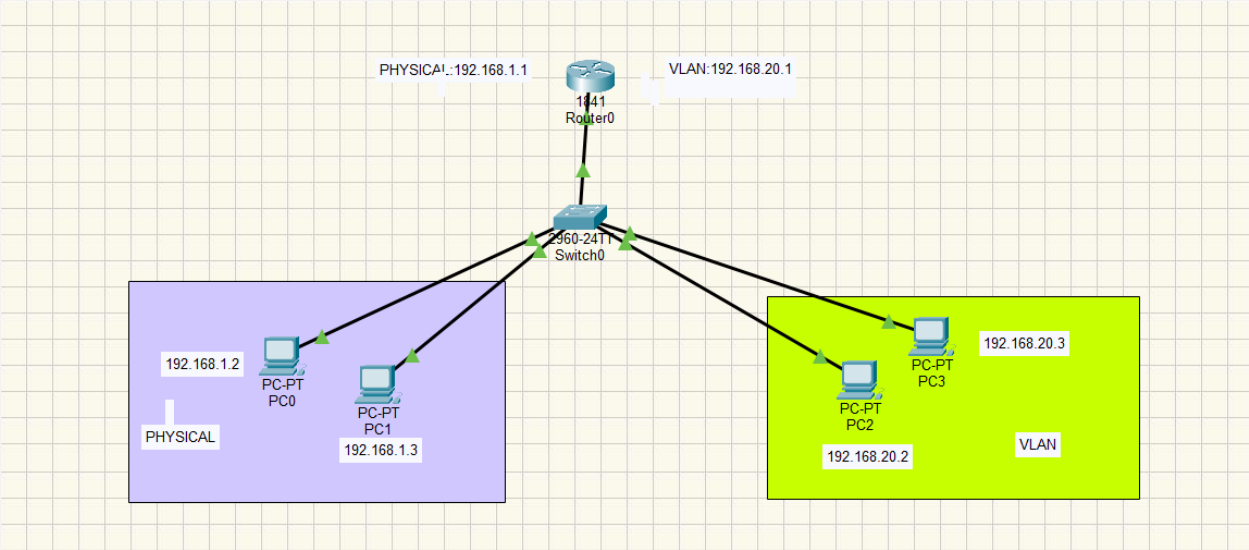


Figure 1: Star topology layout in Cisco Packet Tracer

Device Configuration

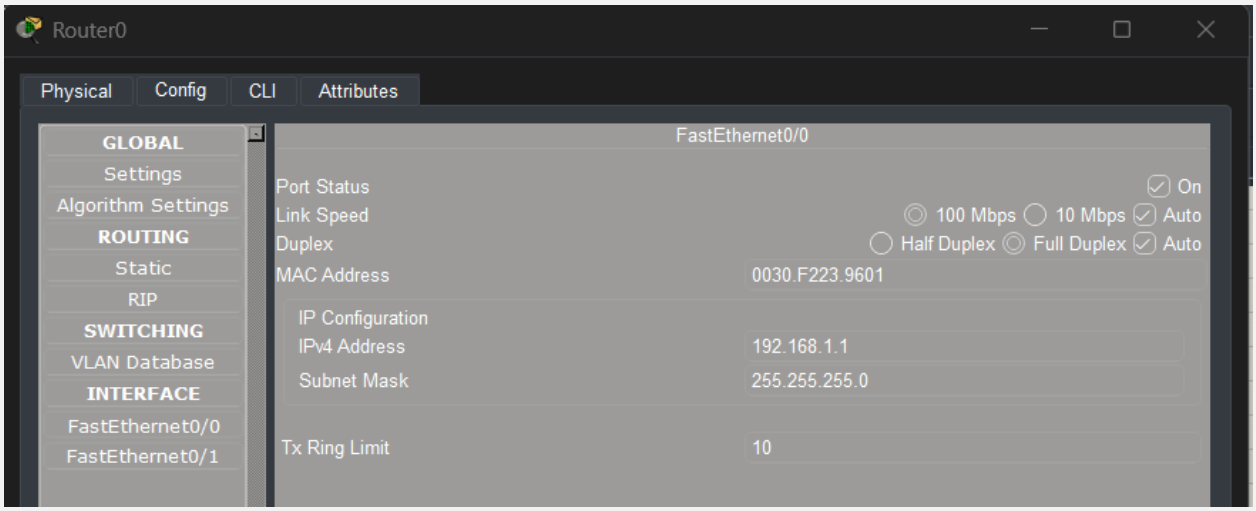


Figure 2.1: Router0 configuration window

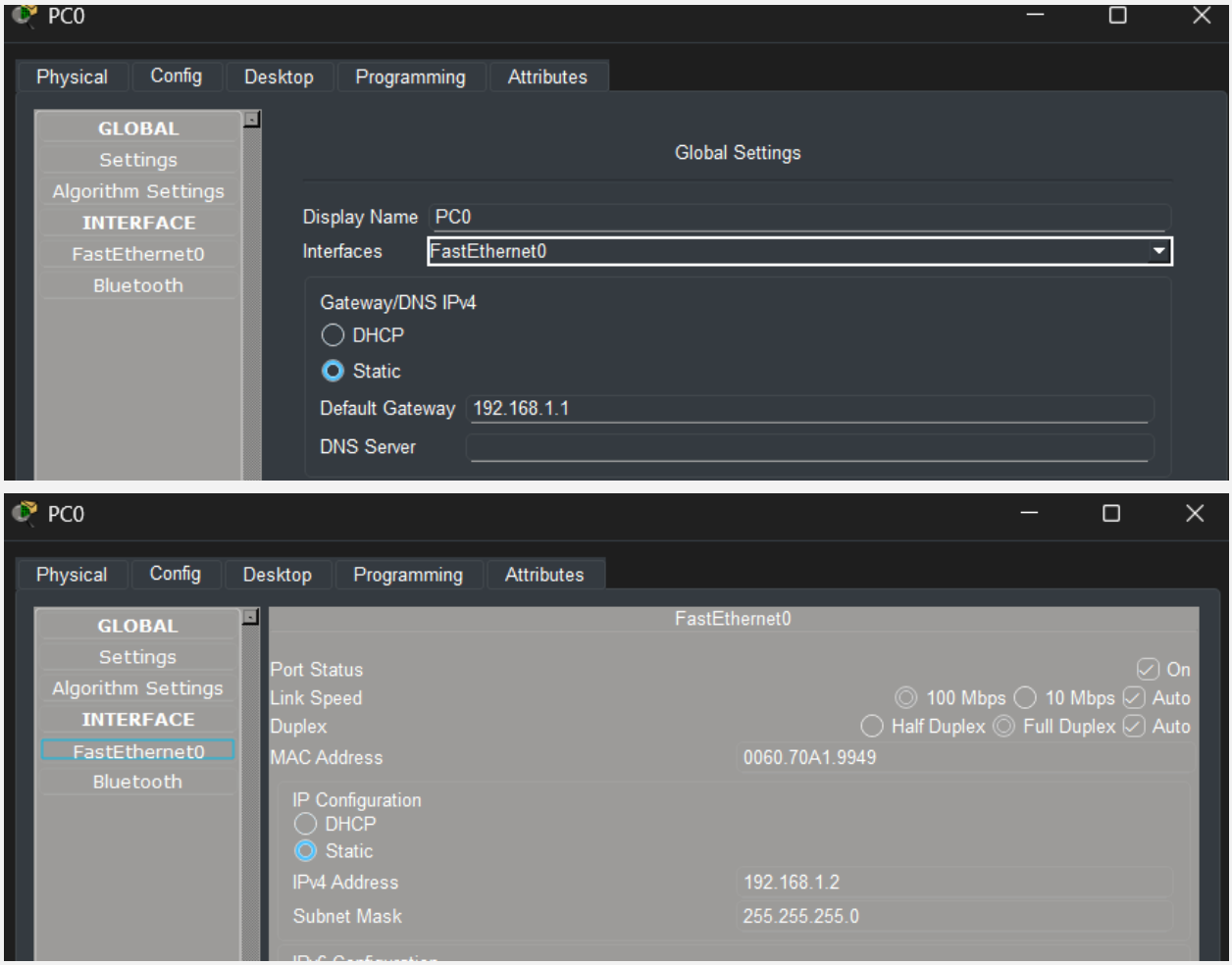


Figure 2.2: PC0 IP address and default gateway configuration

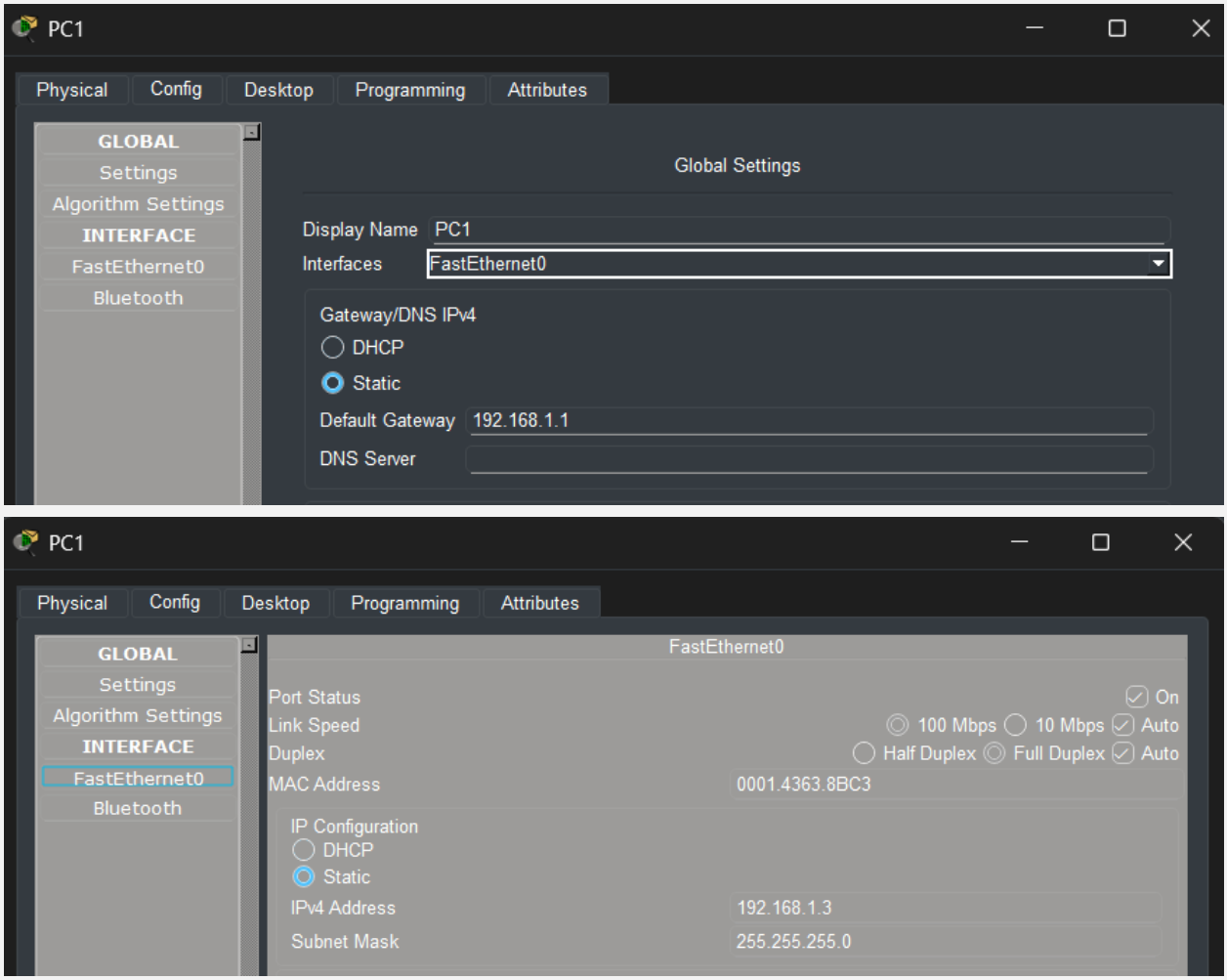


Figure 2.3: PC1 IP address and default gateway configuration

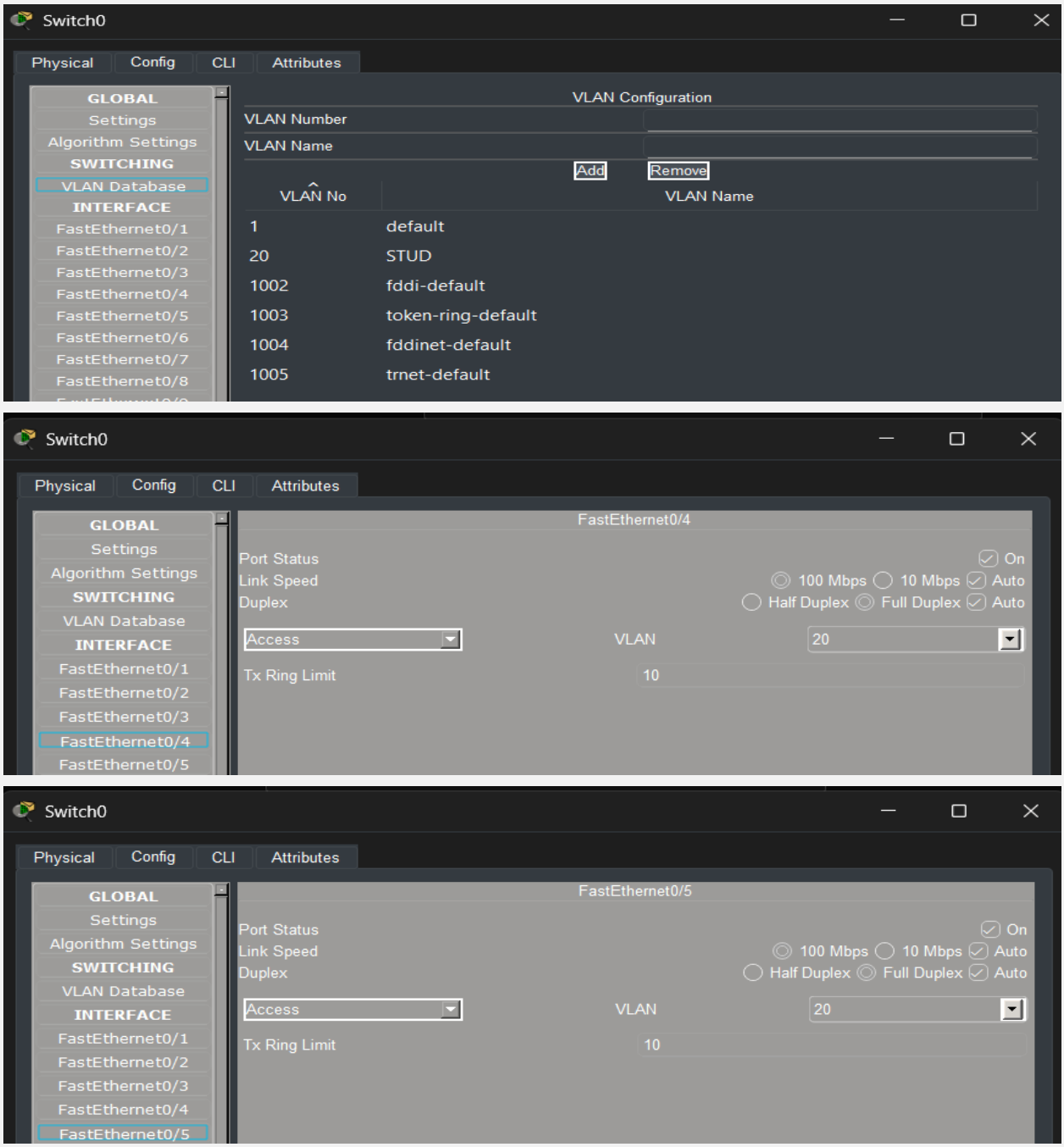


Figure 2.4: VLAN 20 (STUD) creation and trunk configuration of FastEthernet 0/4 and 0/5 ports

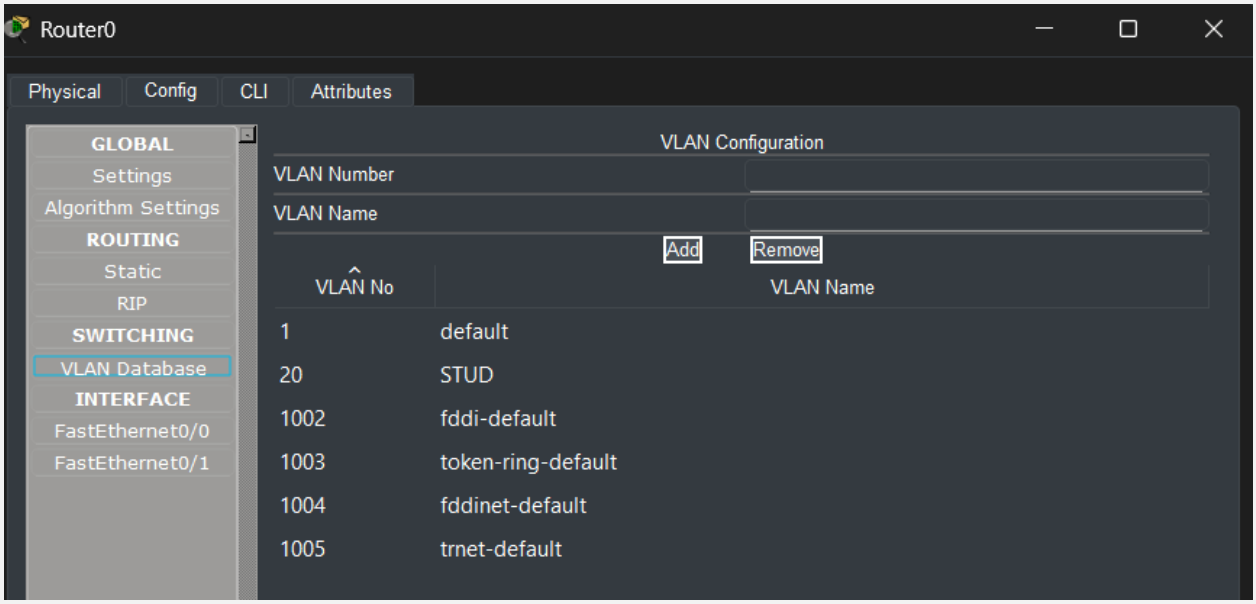




Figure 2.5: VLAN configuration on Router0

Connectivity Test (Ping / Simulation Result)

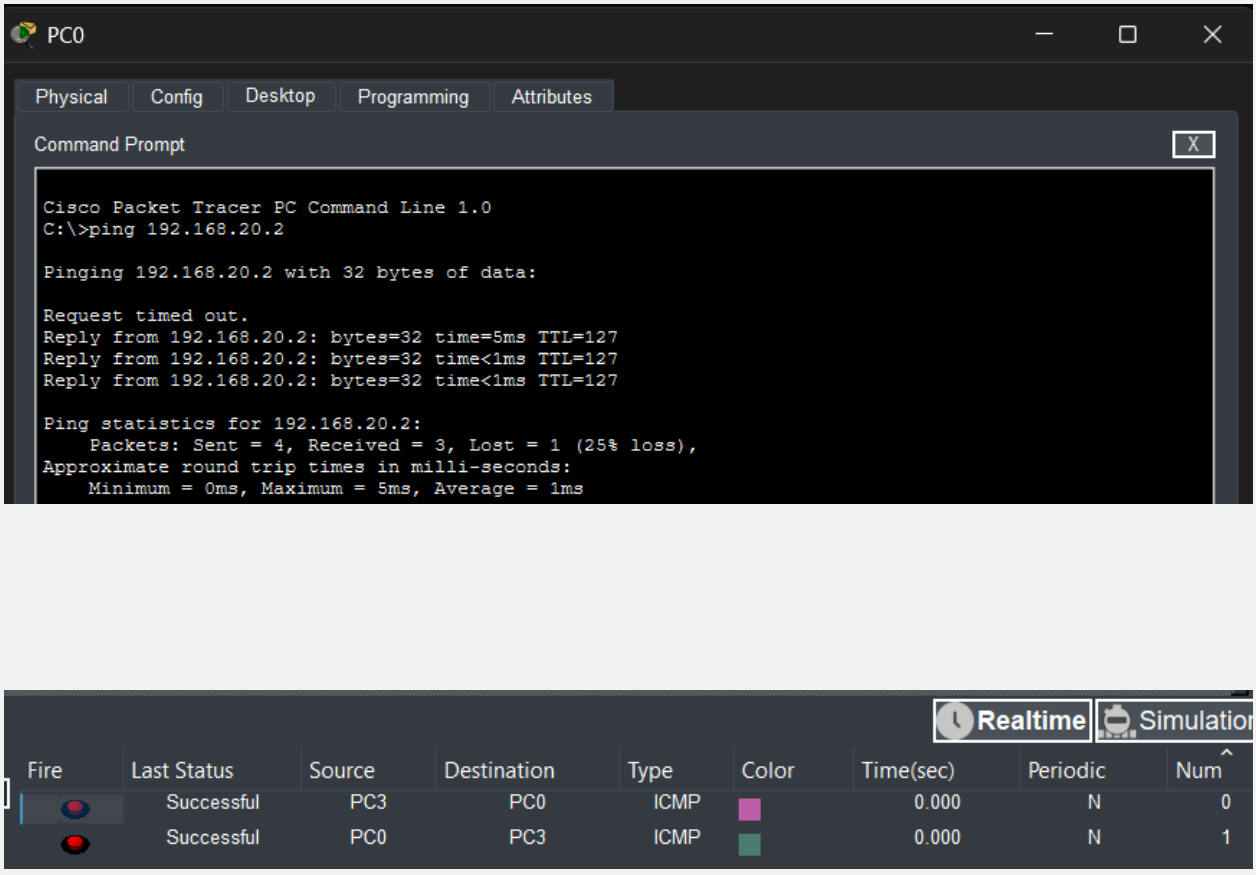


Figure 3: Successful ping / packet simulation result between PCs

Process of Designing the Network in Packet Tracer

I. Creating the Basic Network Topology
   a. The network was designed in Cisco Packet Tracer using one router, one switch, and four PCs.
   b. The router was placed at the top of the workspace, followed by the switch, with the four PCs placed below the switch.
   c. The two PCs on the left were used as the physical LAN computers, while the two PCs on the right were used as the VLAN computers.
   d. The Router0 was connected to the switch using a Copper Straight-Through cable.
   e. All four PCs were connected to the switch using Copper Straight-Through cables.
   f. The completed basic topology consisted of the router connected to the switch and the switch connected to all four PCs.

II. Configuring the Physical LAN PCs
   a. The two PCs on the left were configured as the computers belonging to the physical LAN.
   b. The IP address and default gateway were assigned to each PC according to the network being used.
   c. The default gateway was configured so that the PCs could communicate with the router and other networks.
   d. The IP addresses were assigned from the same network as the gateway so that the PCs could communicate within the physical LAN.
   e. The IP configuration was checked on each PC to ensure that the assigned IP address, subnet mask, and gateway were correct.

III. Configuring the VLAN PCs
   a. The two PCs on the right were selected to represent the computers belonging to the VLAN.
   b. These PCs were configured using the 192.168.20.0/24 network.
   c. The default gateway for the VLAN PCs was set to 192.168.20.1.
   d. The first VLAN PC was assigned the IP address 192.168.20.2 with the subnet mask 255.255.255.0.
   e. The second VLAN PC was assigned another available IP address from the same network, with the same subnet mask and default gateway.
   f. This ensured that both PCs were part of the same logical VLAN network.

IV. Creating the VLAN on the Switch
   a. The switch was opened and the Config tab was selected.
   b. The VLAN Database section was opened to create a new VLAN.
   c. A new VLAN was created with VLAN ID 20 and the name STUD.
   d. The VLAN was added to the switch database.
   e. VLAN 20 was therefore created as a separate logical network within the switch.

V. Configuring the Trunk Ports
   a. The connection between the switch and the router was identified.

b. The switch ports connected to the router were identified as FastEthernet 0/4 and FastEthernet 0/5.
c. These ports were configured as trunk ports.
d. Trunking allows VLAN traffic to travel between the switch and router through the configured connection.
e. The trunk configuration is required so that the router can receive and process traffic belonging to VLAN 20.

VI. Assigning the VLAN to the Required Switch Ports

a. The ports connected to the two VLAN PCs were identified on the switch.
b. These ports were configured as access ports and assigned to VLAN 20 (STUD).
c. This made the two PCs members of VLAN 20.
d. The remaining PCs continued to operate on the physical LAN network.
e. The port assignments were checked to ensure that the correct PCs were connected to the correct VLAN.

VII. Configuring VLAN on Router0

a. After configuring VLAN 20 on the switch, the router was configured to provide communication between the VLAN and the rest of the network.
b. The Command Line Interface (CLI) of Router0 was opened.
c. Configuration mode was entered using:
config t
d. The required router subinterface was selected using:
int f0/0.1
e. VLAN encapsulation was configured using:
encapsulation dot1q 20
f. The gateway IP address for VLAN 20 was assigned:
ip address 192.168.20.1 255.255.255.0
g. The configuration was then exited.
h. The router now acts as the default gateway for the devices belonging to VLAN 20.

VIII. Checking the Network Connections

a. After completing the device, VLAN, trunk, and router configurations, all physical connections were checked.
b. The connection indicators in Cisco Packet Tracer were verified to ensure that the links between the router, switch, and PCs were active.
c. The IP addresses, subnet masks, and default gateways of the PCs were checked once again to make sure there were no configuration errors.

IX. Testing Connectivity

a. The connectivity between the network devices was tested using the ping command from the PCs.
b. A PC was opened and Command Prompt was selected.
c. The IP address of another network device was entered with the ping command.

X. For example:

ping 192.168.20.2

a. Successful reply from messages confirmed that packets were being transferred successfully.
b. Connectivity to the default gateway 192.168.20.1 was also checked.
c. The successful ping test verified that the IP addressing, VLAN assignment, trunk configuration, and router configuration were functioning correctly.

XI. Final Network

a. After completing all the configurations and connectivity tests, the network successfully consisted of a physical LAN and VLAN 20 (STUD).
b. The switch was responsible for connecting the PCs and separating the VLAN traffic, while Router0 provided the required gateway and routing functionality.
c. The final connectivity test confirmed that the configured network was operating successfully in Cisco Packet Tracer.

Conclusion

The VLAN was successfully designed and configured using **Cisco Packet Tracer**, demonstrating how a physical network can be logically divided into separate broadcast domains. The experiment involved configuring **IP addresses, VLAN 20 (STUD), switch ports, trunk connections, and router subinterfaces** to enable communication between the connected networks. Connectivity was then verified using the **ping command**. This experiment helped reinforce the practical understanding of **VLAN configuration, network segmentation, trunking, and router-based communication**.